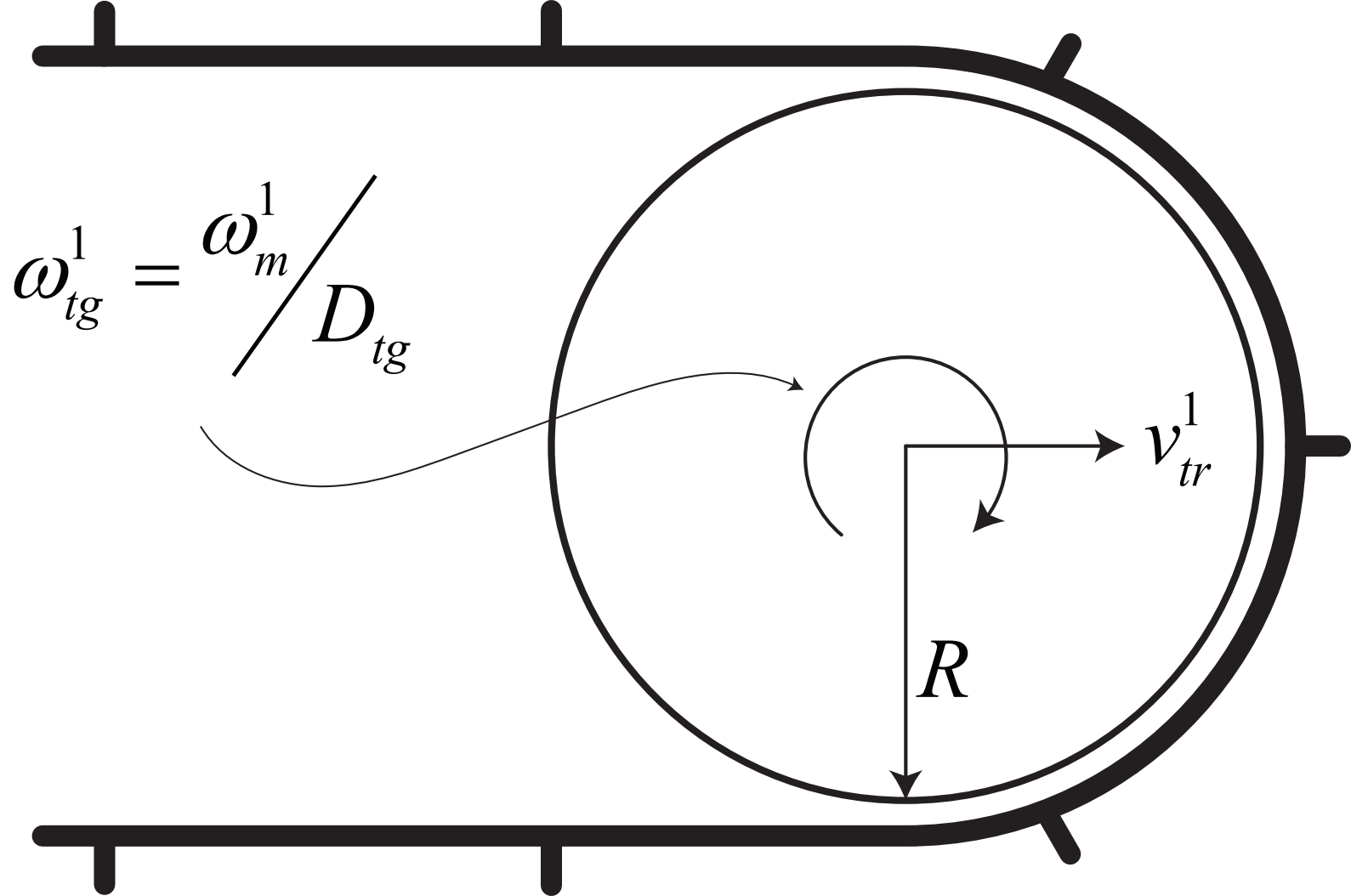
$\omega_{tg}^{1} = \omega_{m}^{1} / D_{tg}$
v_{tr}^{1}
R